Use-Case Flow Specification

Use Case ID: UC-01

Use Case Name: Log Maintenance Ticket

Primary Actor: Hostel Resident

Goal / Brief Description: Allows a hostel resident to report a plumbing, electrical, or carpentry issue so it is recorded, geo-tagged, and routed to the Maintenance Warden for dispatch.

Preconditions

- The Resident is authenticated and logged into the ticketing system.
- The Resident's room/hostel-block assignment is registered in the system.

Postconditions

- Success: A new ticket exists with a unique ID, status “Open,” geo-tag recorded, and it is visible in the Maintenance Warden's queue.
- Failure: No ticket record is created and the Resident's submitted data is not persisted.

Main Success Scenario

1. Hostel Resident selects “Report New Issue” in the ticketing app.
2. System displays the ticket form (room number, issue category, severity, photo upload).
3. Resident enters the room number, selects an issue category (Plumbing / Electrical / Carpentry) and severity, and attaches at least one photo.
4. Resident submits the ticket.
5. System validates the submitted details, including the include use case «Validate Ticket Details» (mandatory room tag and category check).
6. System generates a unique Ticket ID, records the geo-location tag, and sets ticket status to “Open.”
7. System routes the ticket to the Maintenance Warden’s queue.
8. System displays a confirmation screen with the Ticket ID to the Resident.
9. Use case ends successfully.

Alternate Flow — A1: Incomplete Submission

Branch point: Step 5, during «Validate Ticket Details».

A1.1. Validation fails because a mandatory field (e.g., room number) is missing or the photo attachment is absent.

A1.2. System rejects the submission and displays an error message naming the missing field(s); no Ticket ID is generated.

A1.3. Resident corrects the missing information and resubmits the form; flow resumes at Main Success Scenario step 5.